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ELECTROLYTIC CELL FOR THE PRODUCTION OF HYPOCHLOROUS ACID

Abstract

An electrolytic cell for the production of hypochlorous acid and sodium hydroxide from brine as separate product streams is described. The electrolytic cell includes an electrolysis chamber containing an anode and a cathode; electrical connections to each of the anode and cathode; a brine inlet into the electrolysis chamber for the supply of a brine solution into the electrolysis chamber; a first liquid outlet from the electrolysis chamber proximal the anode; a second liquid outlet from the electrolysis chamber proximal the cathode; at least one pump arranged to draw liquid from the electrolysis chamber through the first liquid outlet at a first flow rate and through the second liquid outlet at a second flow rate. The cell generates hypochlorous acid for use in situ, for example as a cleaning product in the home or at commercial or industrial premises.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present invention relates to an electrolytic cell which produces hypochlorous acid as a product. The invention also relates to method of producing hypochlorous acid using such cells, along with hypochlorous acid generators and household cleaning product dispensers containing such cells.

BACKGROUND OF THE INVENTION

[0002] Hypochlorous acid, HOCl, can be prepared through the electrolysis of brine (concentrated sodium chloride solution). During electrolysis, chloride ions are oxidised at the anode to form chlorine and water molecules are reduced at the cathode to form sodium hydroxide and hydrogen. The chlorine formed at the anode reacts with water to form hypochlorous acid.

[0003] During the commercial production of sodium hypochlorite (household bleach) through electrolysis, the hypochlorous acid formed at the anode reacts with the sodium hydroxide produced at the cathode to form the sodium hypochlorite product, NaOCl. This process is performed on a mega-tonne scale and is known as the chlor-alkali process.

[0004] Sodium hypochlorite has the advantage of being relatively stable such that it can be stored for long periods without degradation or loss of biocidal activity.

[0005] Hypochlorous acid has a biocidal activity around 100 times that of sodium hypochlorite. As a result, a much smaller quantity of hypochlorous acid is required to achieve a biocidal effect equivalent to sodium hypochlorite bleach and a reduced contact time with surfaces is required for an effective kill rate. This would be a significant advantage for household cleaning products which are typically “quick-use”, with consumers applying the products to surfaces for a short period before rinsing. Unfortunately hypochlorous acid degrades quickly during storage, losing its biocidal activity. It is therefore not possible to prepare hypochlorous acid commercially for supply to consumers to use as an alternative to ordinary bleach.

[0006] There is a need for products and processes which allow a consumer to harness the increased biocidal activity of hypochlorous acid to provide improved household cleaning products. There is also a need for more convenient and versatile household devices which provide consumers with a choice of cleaning product which is tailored to the intended application.

SUMMARY OF THE INVENTION

[0007] A first aspect of the invention is an electrolytic cell for the production of hypochlorous acid and sodium hydroxide from brine as separate product streams, the electrolytic cell comprising:

[0008] an electrolysis chamber comprising an anode and a cathode; [0009] electrical connections to each of the anode and cathode; [0010] a brine inlet into the electrolysis chamber for the supply of a brine solution into the electrolysis chamber; [0011] a first liquid outlet from the electrolysis chamber proximal the anode; [0012] a second liquid outlet from the electrolysis chamber proximal the cathode; and [0013] at least one pump arranged to draw liquid from the electrolysis chamber

through the first liquid outlet at a first flow rate and through the second liquid outlet at a second flow rate; [0014] wherein either: [0015] (a) the first flow rate and second flow rate are variable relative to one another; or [0016] (b) the first flow rate and second flow rate are fixed relative to one another at a predetermined difference in flow rate.

[0017] The cell generates hypochlorous acid for use in situ, for example as a cleaning product in the home or at commercial or industrial premises. In this way, prolonged storage of the hypochlorous acid is unnecessary and the product can be used immediately or stored for a short period of time before use, avoiding any loss of biocidal activity. Furthermore, the cell provides a convenient means for a user to generate a biocidal sanitising product much more potent than ordinary bleach on-demand, from freely available salt solution which may be prepared by mixing ordinary table salt into water.

[0018] Not only does the cell generate hypochlorous acid from the first liquid outlet, but the presence of a separate second liquid outlet proximal the cathode provides a source of sodium hydroxide solution which has separate utility as a cleaning product with particular effectiveness as a degreaser (sodium hydroxide is a common constituent of oven cleaner). The cell therefore dispenses two separate products which have different cleaning and sanitising applications, providing a convenient and versatile source of on-demand cleaning products.

[0019] Surprisingly, the inventors have found that by providing first and second flow rates which are variable relative to one another, the nature of the products from the cell can be tailored according to the desire of the user and cell efficiency improvements are provided. By adjusting the relative flow rates, it has been found that it is possible to tailor the pH of the product which passes out of the first liquid outlet of the cell. In this way, the solution leaving the first liquid outlet can have properties tailored for use in a desired application. For example, a solution of hypochlorous acid of very low pH would find use as a limescale remover. A slight increase in the pH of the solution would make the solution more suitable for use as a general sanitiser, for example for sanitising floors and work surfaces. A further increase in pH towards the neutral range would provide a solution more appropriate for use in sanitising delicate surfaces.

[0020] Similarly, by providing first and second flow rates which are fixed relative to one another, but at a predetermined difference in flow rate, it is possible for either the manufacturer or end user to select a fixed flow bias to tailor the nature of the products passing out of the cell and provide consistent product generation.

[0021] The invention therefore provides a device which allows a user to precisely control the nature of the product generated, which can be tailored to a specific desired cleaning or sanitising application.

[0022] A second aspect of the invention is a hypochlorous acid generator comprising an electrolytic cell according to the first aspect, further comprising: [0023] one or more of a brine reservoir and a brine feed conduit; [0024] a hypochlorous acid storage unit; [0025] a sodium hydroxide storage unit; [0026] a power source; and [0027] a product dispensing device.

[0028] Such a generator provides a means to produce hypochlorous acid on-demand, for example in a home or workplace, from freely available salt solution precursor which is stored in the brine reservoir. This provides a convenient and versatile means to dispense hypochlorous acid for use as a biocide along with other useful products, such as sodium hydroxide which finds use as a degreaser.

[0029] A third aspect of the invention provides a method of producing hypochlorous acid using a cell according to the first aspect, the method comprising: [0030] passing brine through the brine inlet and into the electrolysis chamber; [0031] applying a voltage across the electrical connections of the anode and cathode to electrolyse the brine; [0032] drawing a first product stream from the first liquid outlet at a first flow rate, the first product stream comprising hypochlorous acid; and [0033] drawing a second product stream from the second liquid outlet at a second flow rate, the second product stream comprising sodium hydroxide; [0034] wherein the concentration of

hypochlorous acid in the first product stream is higher than the concentration of hypochlorous acid in the second product stream.

[0035] The method of the third aspect provides a convenient and versatile method to produce hypochlorous acid for use as a biocide (sanitiser) along with other useful products, such as sodium hydroxide which finds use as a degreaser.

[0036] A fourth aspect of the invention provides a household cleaning product dispenser comprising the electrolytic cell according to the first aspect. In this way, a consumer is able to dispense a desired cleaning product on-demand which eliminates the need to separately purchase cleaning products for use in the home or workplace. In addition, the superior biocidal activity of hypochlorous acid relative to sodium hypochlorite bleach is made available to the ordinary consumer, since there is no need to store unstable hypochlorous acid for long periods due to its in-situ generation.

[0037] Preferred and/or optional features of the invention will now be set out. Any aspect of the invention may be combined with any other aspect of the invention unless the context demands otherwise. Any of the preferred and/or optional features of any aspect may be combined, either singly or in combination, with any aspect of the invention unless the context demands otherwise.

[0038] The term “electrolytic cell” refers to an arrangement of an anode and a cathode, separated by an electrolyte and adapted to function as a brine electrolyser when a suitable voltage is applied across the anode and cathode.

[0039] “Hypochlorous acid” refers to the compound HOCl, or its partially or fully dissociated form in solution where it may exist as the dissociated ions H^+ and ClO^- .

[0040] “Sodium hydroxide” refers to the compound NaOH, or its partially or fully dissociated form in solution where it may exist as the dissociated ions Na^+ and OH^- .

[0041] The term “electrolysis chamber” refers to a region or volume within an electrolyser which contains the anode and the cathode and within which decomposition of chemical species at the anode and cathode occurs when a suitable voltage is applied across the anode and cathode.

[0042] The term “brine” refers to a solution of sodium chloride (NaCl) in water, at any concentration.

[0043] The first aspect of the invention is an electrolytic cell for the production of hypochlorous acid and sodium hydroxide from brine as separate product streams as defined in claim 1.

[0044] In some embodiments, the electrolytic cell is for the production of hypochlorous acid and sodium hydroxide as separate products. In some embodiments, the electrolytic cell is for the production of a first product comprising a mixture of hypochlorous acid and hydrochloric acid, and a second separate product comprising sodium hydroxide.

[0045] In some embodiments, the at least one pump comprises: [0046] a first pump arranged to draw liquid from the electrolysis chamber through the first liquid outlet at the first flow rate; and [0047] a second pump arranged to draw liquid from the electrolysis chamber through the second liquid outlet at the second flow rate.

[0048] Providing separate first and second pumps provides a simple and reliable way to control the flow rates at the first and second liquid outlets from the cell. However, the skilled person will appreciate that there may be alternative ways to achieve the first and second flow rates without requiring two separate pumps. For example, a single pump may be used along with two separate valves, wherein the operation of the valves may be controlled independently of one another in order to vary the first and second flow rates.

[0049] The at least one pump preferably comprises one or more peristaltic pumps. In some embodiments, the at least one pump preferably consists of one or more peristaltic pumps. In some embodiments, the at least one pump consists of two peristaltic pumps, a first peristaltic pump arranged to draw liquid from the electrolysis chamber through the first liquid outlet at the first flow rate and a second peristaltic pump arranged to draw liquid from the electrolysis chamber through the second liquid outlet at the second flow rate. One or both of the first and second peristaltic

pumps are controllable in order to vary the flow rates.

[0050] The at least one pump may be connected with one or more pump motors.

[0051] In some embodiments, the one or more pump motors are powered by a power source, for example an electric battery, via a pulse width modulation (PWM) module.

[0052] The peristaltic pumps may comprise a peristaltic pump head connected with a peristaltic pump motor. In some embodiments, the first and second peristaltic pumps are each connected to and controlled by the same motor. In some embodiments, the first and second peristaltic pumps are connected to and controlled by different motors. Each peristaltic pump motor may be powered by an electric battery via a pulse width modulation (PWM) module.

[0053] In some embodiments, the cell does not comprise any pumps at a location upstream of the cell. In some embodiments, the cell does not comprise any pumps at a location upstream of the first and second liquid outlets. In other words, the flow of liquid through the cell is entirely achieved by “pulling” liquid through the cell and out of the liquid outlets using the one or more downstream pumps, preferably two separate downstream pumps, without any pump being present which “pushes” liquid through the cell from upstream. It has been found that this arrangement provides more precise control over the flow of liquid through the cell and facilitates the achievement of a flow imbalance between the first and second liquid outlets. By separately “pulling” liquid from each outlet without any “pushing”, the flow rate at each outlet can be fine-tuned, thereby permitting fine-tuning of the composition of the electrolysis products leaving the cell.

[0054] Furthermore, gas bubbles may be generated within the cell during electrolysis, and these are likely to interfere with the electrolysis process leading to unwanted variations in the pH of the product and/or the concentration of hypochlorous acid within the product. By “pulling” the solution in this way using a downstream pump, for example a downstream peristaltic pump, more effective and efficient gas bubble removal from the cell is achieved which would not be possible using upstream pumping. This is particularly important at the second liquid outlet, since the cathode in general produces higher levels of insoluble gases during electrolysis.

[0055] In the electrolytic cell, a first liquid outlet from the electrolysis chamber is proximal the anode and a second liquid outlet from the electrolysis chamber proximal the cathode. In this way, the products dispensed from the first liquid outlet and the second liquid outlet are specific to the products produced by electrolysis at the particular electrode proximal to that outlet, and differ from one another, because products of electrolysis at the anode are drawn off at the first liquid outlet and products of electrolysis at the cathode are drawn off at the second liquid outlet, before those products are able to mix with one another to any significant extent within the cell.

[0056] The first liquid outlet from the electrolysis chamber is proximal the anode. In other words, the first liquid outlet is adapted to (for example, positioned to) primarily draw liquid from the vicinity of the anode, or is adapted to (for example, positioned to) preferentially draw liquid from the area proximal the anode relative to the area proximal the cathode.

[0057] In this way, the first liquid outlet functions to dispense the electrolysis products formed at the anode, i.e. primarily a solution of hypochlorous acid.

[0058] In some embodiments the first liquid outlet from the electrolysis chamber is located at a position which is closer to at least a part of the anode than it is to any part of the cathode. In this way the physical location of the first liquid outlet within the cell provides preferential extraction of products from the vicinity of the anode.

[0059] The second liquid outlet from the electrolysis chamber is proximal the cathode. In other words, the second outlet is configured to (for example, positioned to) primarily draw liquid from the vicinity of the cathode, or is configured to (for example, positioned to) preferentially draw liquid from the area proximal the cathode relative to the area proximal the anode.

[0060] In this way, the second liquid outlet functions to dispense the electrolysis products formed at the cathode, i.e. primarily a solution of sodium hydroxide.

[0061] In some embodiments the second liquid outlet from the electrolysis chamber is located at a

position which is closer to at least a part of the cathode than it is to any part of the anode. In this way the physical location of the second liquid outlet within the cell provides preferential extraction of products from the vicinity of the cathode.

[0062] The first and second liquid outlets may each be located within an internal wall of the electrolysis chamber, the electrolysis chamber being otherwise liquid-impermeable (with the exception of the brine inlet) such that the only permitted passage of liquid out of the cell is through either the first or second liquid outlet.

[0063] The first liquid outlet may have an area of at least 1.5 mm.^{sup.2}. The first liquid outlet may have an area of up to 10.0 mm.^{sup.2}. In some embodiments, the first liquid outlet has an area of 1.5 to 10.0 mm.^{sup.2}

[0064] In some embodiments, the inner wall of the electrolysis chamber adjacent to the first liquid outlet is angled or funnelled towards the first liquid outlet. This facilitates the flow of liquid towards and into the first liquid outlet, establishing flow conditions within the cell which further discourage mixing of the anode electrolysis products with the cathode electrolysis products. This arrangement also aids the efficient removal of gas bubbles from the cell.

[0065] In some embodiments, a portion of the inner wall of the electrolysis chamber adjacent to the first liquid outlet is angled or funnelled towards the first liquid outlet, wherein the length of the portion of inner wall which is angled or funnelled is at least twice the diameter of the first liquid outlet, for example at least three times or at least four times the diameter of the first liquid outlet. Providing such a length of wall which is angled or funnelled provides further improvements in the flow towards the outlet.

[0066] In some embodiments, the inner wall of the electrolysis chamber adjacent to the second liquid outlet is angled or funnelled towards the second liquid outlet. This facilitates the flow of liquid towards and into the second liquid outlet, establishing flow conditions within the cell which further discourage mixing of the anode electrolysis products with the cathode electrolysis products. This arrangement also aids the efficient removal of gas bubbles from the cell.

[0067] In some embodiments, a portion of the inner wall of the electrolysis chamber adjacent to the second liquid outlet is angled or funnelled towards the second liquid outlet, wherein the length of the portion of inner wall which is angled or funnelled is at least twice the diameter of the second liquid outlet, for example at least three times or at least four times the diameter of the second liquid outlet. Providing such a length of wall which is angled or funnelled provides further improvements in the flow towards the outlet.

[0068] In some embodiments, each of the anode and cathode are liquid-permeable. In this way, the anode and cathode are adapted to permit the flow of liquid through them, which may facilitate the operation of the cell and provide improved flow of brine and electrolysis products within the electrolysis chamber.

[0069] In some embodiments, each of the anode and cathode are a liquid-permeable mesh.

[0070] In other embodiments, each of the anode and cathode may be liquid-impermeable. In such embodiments means for the passage of liquid past or around the electrode will be provided.

[0071] For example, there may be a gap or channel around the periphery of the electrode to permit the passage of liquid past the electrode and towards the appropriate liquid outlet.

[0072] It would be desirable to provide a lightweight, hand-held hypochlorous acid generating device containing the electrolytic cell of the first aspect. However in order to achieve this and enable sufficiently long run time, the power draw of the device must remain low. In some embodiments, the cell is adapted to generate useful quantities of hypochlorous acid at low power. To achieve this, in some embodiments, the distance between the anode and the cathode is reduced. For example, in some embodiments the separation between the anode and the cathode is less than 3.0 mm, for example less than 2.0 mm. Providing a closer separation of anode and cathode reduces the internal resistance of the cell, reducing power requirements. Alternatively or additionally, in some embodiments the concentration of sodium chloride in the brine solution fed to the cell is

increased, as explained in more detail below. A higher concentration of sodium chloride reduces the internal resistance of the cell during electrolysis, reducing power requirements. Alternatively or additionally, in some embodiments the properties of the anode and/or the cathode may be adapted to reduce the power draw of the cell. For example, as explained below, the anode and cathode may comprise a catalyst, such as an $\text{IrO}_2/\text{RuO}_2$ mixed metal oxide catalyst, to increase the rate of electrolysis allowing a given concentration of hypochlorous acid to be generated at lower power requirements. One or more of these adaptations may be used within the cell in order to reduce the power draw and ensure adequate run time when a battery power source is used.

[0073] In some embodiments, one or both of the anode and cathode comprise a metal substrate and a noble metal oxide catalyst supported on the substrate. In some embodiments, one or both of the anode and cathode comprise a transition metal substrate and a platinum-group metal oxide catalyst supported on the substrate.

[0074] In some embodiments, the metal substrate comprises titanium or platinum, and the noble metal oxide catalyst comprises an $\text{IrO}_2/\text{RuO}_2$ mixed metal oxide catalyst. In some embodiments, the metal substrate comprises titanium, and the noble metal oxide catalyst comprises an $\text{IrO}_2/\text{RuO}_2$ mixed metal oxide catalyst.

[0075] In some embodiments, the anode and cathode each comprise a catalyst selected from IrO_2 , RuO_2 , Pt, boron-doped diamond (BDD) and mixtures thereof. In some embodiments, the anode and cathode each comprise a catalyst selected from IrO_2 , RuO_2 , Pt and mixtures thereof.

[0076] In some embodiments, the catalyst supported on the anode is the same as the catalyst supported on the cathode. This facilitates polarity switching, after which the cathode will become the anode and vice versa.

[0077] The anode and the cathode may each comprise a substantially planar film or sheet. In some embodiments the film or sheet defines a plurality of apertures for the passage of liquid from one side of the sheet to the other, through the plane of the sheet. In some embodiments each of the anode and cathode have a porosity of from 80% to 40%, where porosity is defined as the total area of the apertures as a percentage of the total area of the sheet. In some embodiments, each of the anode and cathode have a porosity of from 75% to 40%, for example from 73% to 45%.

[0078] In some embodiments, the anode comprises a planar sheet having a thickness of from 0.15 mm to 1.50 mm.

[0079] In some embodiments, the cathode comprises a planar sheet having a thickness of from 0.15 mm to 1.50 mm.

[0080] Each of the anode and cathode may independently have a length of at least 20 mm, for example at least 25 mm. Each of the anode and cathode may independently have a length of up to 120 mm, for example up to 100 mm.

[0081] Each of the anode and cathode may independently have a width of at least 15 mm, for example at least 18 mm. Each of the anode and cathode may independently have a width of up to 80 mm, for example up to 75 mm.

[0082] In some embodiments, the anode and cathode are each independently planar sheets having a length of 25 to 100 mm and a width of 18 to 75 mm. In some embodiments, the anode and cathode are each independently planar sheets having a length of 30 to 100 mm and a width of 20 to 75 mm.

[0083] In some embodiments, the anode and cathode are each planar sheets and lie substantially parallel to one another within the cell. The separation between the anode and the cathode may be from 1.5 mm to 5 mm.

[0084] In some embodiments, the anode and the first liquid outlet are arranged such that at least a portion of the liquid passing from the brine inlet to the first liquid outlet must pass through the anode, for example through one or more apertures in the porous anode. This ensures that at least a portion of the liquid entering the cell is directed towards and ultimately through the anode, facilitating the oxidation of chloride ions at the anode.

[0085] In some embodiments, the cathode and the second liquid outlet are arranged such that at least a portion of the liquid passing from the brine inlet to the second liquid outlet must pass through the cathode, for example through one or more apertures in the porous cathode.

[0086] This ensures that at least a portion of the liquid entering the cell is directed towards and ultimately through the cathode, facilitating the reduction of water at the anode.

[0087] In some embodiments, the anode is a porous anode stretching across the entire electrolysis chamber such that liquid must pass through the anode to reach the first liquid outlet.

[0088] In some embodiments, the cathode is a porous cathode stretching across the entire electrolysis chamber such that liquid must pass through the cathode to reach the second liquid outlet.

[0089] In some embodiments, the cell comprises a housing defining the electrolysis chamber and containing the anode and cathode.

[0090] In some embodiments, the housing comprises an opening in fluid communication with the brine inlet and the electrolysis chamber, such that when the housing is at least partially submerged in a brine reservoir, brine from the reservoir enters the housing through the opening and is drawn through the cell by operation of the at least one pump to draw liquid from the electrolysis chamber.

[0091] In this way the cell may be used with an “open” brine reservoir by simply placing the cell into the reservoir such that the housing is at least partially submerged in a brine reservoir (i.e. such that the opening within the housing is submerged in the reservoir). This obviates the need for tubing or piping for conveying brine from a brine reservoir to the cell, thereby providing apparatus which operates in a more straightforward manner and is easier for a user to maintain. A consumer simply places the cell housing into a brine reservoir in order for the cell to function and can easily clean the housing opening when necessary. There is no need for the user to monitor the condition of any brine piping or to replace such piping after failure or breakage. Furthermore, the risk of leakage of brine from the apparatus is reduced or removed.

[0092] In some embodiments, the housing comprises a base portion which comprises a planar external surface, such that the housing may be self-supporting when placed upon a surface, with the planar base portion resting upon that surface. For example, the housing may be self-supporting when resting within a brine reservoir upon a floor of the reservoir. In this way the housing may maintain a stable position for long periods when placed upon a surface, for example within a brine reservoir.

[0093] The base portion may occupy a part of the housing opposite the first liquid outlet, second liquid outlet, first electrical connection and second electrical connection. In this way, when the housing is placed within a brine reservoir upon its base portion, the first liquid outlet, second liquid outlet, first electrical connection and second electrical connection all lie at an upper part of the cell and housing, outside the reservoir. This provides straightforward extraction of products from the outlets and also ensures that the first electrical connection and second electrical connection do not come into contact with the brine reservoir, which could cause a short-circuit or damage the cell.

[0094] In some embodiments, the cell comprises a porous diaphragm arranged between the anode and the cathode. It has been found that the presence of such a diaphragm, although not essential to the functioning of the cell, further discourages the mixing of electrolysis products within the cell, thereby contributing to the generation of distinct products at the first liquid outlet and second liquid outlet. Furthermore, the diaphragm reduces the diffusion of gas across the cell, for example hydrogen gas from the cathode to the anode. The diffusion of gas could push more NaOH from the cathode side to the anode side raising the pH to a level which may not favour the generation of HOCl as the chlorine species at the anode.

[0095] In some embodiments, the diaphragm comprises woven cloth material.

[0096] In some embodiments, the diaphragm comprises a non-hydrophobic or a hydrophilic material. In some cases this may help improve the performance of the diaphragm within the cell. In some embodiments, the diaphragm does not comprise PTFE. The presence of hydrophobic

materials such as PTFE in the diaphragm may reduce the water-permeability of the diaphragm and impact cell performance.

[0097] In some embodiments, the diaphragm is located in contact with the cathode. The diaphragm may be connected with, for example clamped to, the cathode. In some embodiments, the diaphragm is located in contact with the electrode which operates as the cathode during normal operation of the cell, i.e. during product generation cycles. This electrode will become an anode during a cleaning cycle after a polarity switch, however when such cleaning cycles are short in duration relative to the product generation cycles it may be beneficial to locate the diaphragm in contact with the electrode which operates as the cathode during normal operation of the cell.

[0098] In some embodiments, the electrolytic cell does not contain any diaphragm structure between the anode and the cathode. Although as explained above the presence of a diaphragm may provide some benefits, it may also be desirable to actively omit such a diaphragm from the cell design. The locations of the first liquid outlet and the second liquid outlet, along with the chosen flow rates at those outlets, may alone provide the necessary separation of anode and cathode electrolysis products and contribute to the generation of distinct products at the first liquid outlet and second liquid outlet. Achieving this without the presence of a diaphragm within the cell provides a cell of simple and cost-effective design requiring reduced maintenance and containing fewer parts which may be otherwise prone to failure.

[0099] The first flow rate at the first liquid outlet and the second flow rate at the second liquid outlet may be variable relative to one another. In other words, the difference between the flow rates at the first and second liquid outlets may be variable. Put another way, when the first flow rate is $F_{\text{sub.1}}$ and the second flow rate is $F_{\text{sub.2}}$, the magnitude of the parameter $(F_{\text{sub.2}} - F_{\text{sub.1}})$ may be variable. This allows the user of the cell to tailor the composition of the products, for example to suit a desired cleaning application. This may be achieved by varying the flow rate at one of the liquid outlets while keeping the flow rate at the other liquid outlet fixed. Alternatively, the flow rates at both of the liquid outlets may be independently variable.

[0100] In some embodiments, the flow rate at the first liquid outlet proximal the anode (first flow rate) is fixed and the flow rate at the second liquid outlet proximal the cathode (second flow rate) is variable. In other words, the cell may be configured to provide a fixed flow rate out of the first liquid outlet.

[0101] In some embodiments, the cell is configured to provide a higher liquid flow rate from the second liquid outlet than the first liquid outlet. In some embodiments, the cell is configured such that the liquid flow rate from the second liquid outlet cannot be lower than the liquid flow rate from the first liquid outlet. For example, the one or more pumps or their associated control systems may be adapted to prevent a user reducing the second flow rate to be lower than the first flow rate.

[0102] In some embodiments, the cell is configured to provide a fixed first flow rate $F_{\text{sub.1}}$ at the first liquid outlet and a variable second flow rate $F_{\text{sub.2}}$ at the second liquid outlet, wherein the second flow rate $F_{\text{sub.2}}$ is adjustable in order to vary the composition of the first product stream drawn from the first liquid outlet, wherein $F_{\text{sub.2}} > F_{\text{sub.1}}$ both before and after the adjustment.

[0103] In some embodiments, the cell is configured such that $F_{\text{sub.2}}/F_{\text{sub.1}}$ takes a value within the range 1.1 to 2.0, for example from 1.1 to 1.5, or from 1.2 to 1.4.

[0104] In some embodiments, the cell is configured such that the first flow rate $F_{\text{sub.1}}$ is fixed at a value of from 2 to 30 mL/min, and the second flow rate $F_{\text{sub.2}}$ is adjustable from a first value in the range 5 to 60 mL/min to a second value in the range 5 to 60 mL/min different from the first value, wherein both the first and second value of the second flow rate $F_{\text{sub.2}}$ are greater than $F_{\text{sub.1}}$ by a factor of between 1.1 and 2.0, for example from 1.1 to 1.5, or from 1.2 to 1.4.

[0105] It has been found that a higher liquid flow rate from the second liquid outlet (i.e. from the proximity of the cathode) provides an anode-cathode flow imbalance that helps to achieve a suitable pH within the cell which favours the production of hypochlorous acid rather than other less useful products such as sodium hypochlorite.

[0106] To ensure that the hypochlorous acid solution produced by the cell is safe and compliant with regulations, the pH of the solution must be within a certain range. To ensure a pH of less than 7 for the hypochlorous acid product from the first liquid outlet, sodium hydroxide product is removed from the cell via the second liquid outlet. The flow imbalance between the first and second liquid outlets can be set at a level which results in product solution of desired pH. A greater flow rate out of the second liquid outlet relative to the first liquid outlet will remove a greater quantity of sodium hydroxide, thereby reducing the pH of the product from the first liquid outlet. The NaOH removed from the second liquid outlet may be used as a cleaning product in its own right, or may simply be disposed of.

[0107] On the other hand, a very low pH of the hypochlorous acid product solution from the first liquid outlet would be undesirable, firstly from a safety perspective and secondly because it would favour the formation of chlorine gas at the anode. To mitigate this and ensure a pH above this level, the removal of hydrogen gas bubbles from the cell may be controlled by controlling the flow rates at the first and second liquid outlets. For example, controlled reintroduction of NaOH into the solution within the cell may be achieved by reducing the flow rate at the second liquid outlet to allow hydrogen gas bubbles to remain within the cell.

[0108] In some embodiments, the anode divides the electrolysis chamber into an inlet chamber on a first side of the anode, adjacent the cathode, and an anode collection chamber on a second side of the anode, wherein the first liquid outlet is located in the anode collection chamber; such that, in use, liquid passes from the brine inlet into the inlet chamber, then through the anode into the anode collection chamber before passing out of the first liquid outlet.

[0109] In some embodiments, the cathode divides the electrolysis chamber into an inlet chamber on a first side of the cathode, adjacent the anode, and a cathode collection chamber on a second side of the cathode, wherein the second liquid outlet is located in the cathode collection chamber; such that, in use, liquid passes from the brine inlet into the inlet chamber, then through the cathode into the cathode collection chamber before passing out of the second liquid outlet.

[0110] In some embodiments, the first flow rate and second flow rate are fixed relative to one another at a predetermined difference in flow rate. In some embodiments, the predetermined difference in flow rate provides a concentration of HOCl at the first liquid outlet which is higher than would be achievable at other flow rate differences.

[0111] In some embodiments, the first flow rate takes a value $F_{sub.1}$ and the second flow rate takes a value $F_{sub.2}$, and $F_{sub.2} > F_{sub.1}$. A second flow rate which is greater than the first flow rate has been observed to provide more optimal conditions for HOCl generation.

[0112] In some embodiments, the first flow rate takes a value $F_{sub.1}$ and the second flow rate takes a value $F_{sub.2}$, and $F_{sub.2}/F_{sub.1}$ takes a value within the range 1.1 to 2.0, for example from 1.1 to 1.5, or from 1.2 to 1.4. Such flow rate differences have been found to provide optimal pH and HOCl generation.

[0113] The second aspect of the invention is a hypochlorous acid generator comprising an electrolytic cell according to the first aspect, further comprising: [0114] one or more of a brine reservoir and a brine feed conduit; [0115] a hypochlorous acid storage unit; [0116] a sodium hydroxide storage unit; [0117] a power source; and [0118] a product dispensing device.

[0119] The generator of the second aspect may be conveniently located in a home or workplace to generate and dispense products used for cleaning and sanitising applications on demand.

[0120] The generator may comprise a brine reservoir. The brine reservoir may be an open-top tank or vessel which is at least partially filled with brine. In some embodiments the brine reservoir is dimensioned to facilitate the insertion and holding of the cell.

[0121] The generator may alternatively or additionally comprise a brine feed conduit. The brine feed conduit may be connected with the brine inlet of the cell and adapted to feed brine from a brine reservoir to the cell. The brine feed conduit may comprise a tube, for example a flexible tube. The brine feed conduit may comprise a flexible plastic tube. In this way, the generator may be

“plumbed-in” to ensure direct supply of brine to the cell.

[0122] It would be particularly desirable to provide a hypochlorous acid generating device which is hand-held and portable for ease of use in different areas of the home or workplace, being powered by a rechargeable battery power source. However in order to meet safety and compliance legislation very good control of the output of a device is required. Such control of the output of hypochlorous acid is typically challenging for a battery-powered device, for example due to the gradual reduction in voltage supplied by a battery as it discharges. Through careful control of the power supplied to the device and control of the flow rates through the first and second liquid outlets, the inventors have addressed this problem.

[0123] Thus in some embodiments the power source of the generator may be a battery, for example a rechargeable battery, for example a lithium-ion battery. This provides a device which is rechargeable and is able to draw power from the battery during use, providing a device which is cordless, lightweight, hand-held and portable. This ensures that the device can be easily moved, for example to facilitate the cleaning of different rooms of a home or workplace.

[0124] The generator may comprise one or more motors. Motors may be provided to drive the one or more pumps of the electrolytic cell. For example, when the electrolytic cell comprises first and second peristaltic pump heads, these may be driven by first and second respective motors which are in turn powered by the power source.

[0125] As explained above, providing battery power for hypochlorous acid generating devices may be difficult because precise control of the hypochlorous acid product is required. To address this issue, in some embodiments, the generator further comprises a pulse width modulation (PWM) module connected to the battery to modify the voltage applied across the anode and cathode of the cell during electrolysis, thereby providing an average voltage which will not drop as the battery discharges. This improves the control and consistency of the production of hypochlorous acid.

[0126] In some embodiments, the generator further comprises a pulse width modulation (PWM) module connected to the one or more pumps of the electrolytic cell. This improves control over the flow rates of products out of the cell, further improving the consistency of the product generated.

[0127] In general, in order to produce a consistent and controlled amount of hypochlorous acid, the hypochlorous acid generator must provide control over (a) the sodium chloride concentration in the brine solution fed to the cell, (b) the current supplied to the electrodes, and (c) the voltage across the electrodes. Thus in some embodiments, the generator comprises means to determine the concentration of sodium chloride in the brine solution fed to the electrolysis cell. Suitable means are known to the skilled person and include, for example, a detector which measures the conductivity of the solution which provides information on the concentration of sodium chloride in a brine solution. In some embodiments, the generator includes means to adjust the concentration of sodium chloride in the brine solution in response to the measured concentration. For example, if the concentration is measured to be above a pre-determined threshold, the brine solution may be diluted with water to reduce the sodium chloride concentration before feeding to the cell.

[0128] In some embodiments, the voltage across the electrodes is controlled by PWM as explained above, to ensure a constant average voltage during electrolysis, improving the consistency of electrolysis. The constant average voltage provides a constant effective current through the cell for improved consistency of product generation. Although the current may drop slightly over time due to e.g. limescale build-up on the electrodes causing increased resistance, this can be addressed by the use of polarity switching to provide a cleaning cycle, and is expected to be a minor effect relative to the current drop which would be experienced in the absence of PWM due to the fall in voltage as the battery discharges.

[0129] In some embodiments, the generator further comprises a means to combine product streams in predetermined proportions. For example, the sodium hydroxide solution and hypochlorous acid solution may be combined in a predetermined proportion to provide a product of predetermined pH or chemical composition thereby providing a product which may be tailored for a specific end use.

[0130] In some embodiments, the hypochlorous acid generator comprises a control unit which facilitates control of one or more of the first flow rate and second flow rate of liquid out of the cell. For example, the control unit may provide means for a user to control one or both of the first flow rate and second flow rate. In some embodiments, the control unit provides means to control only the second flow rate. In some embodiments the first flow rate is fixed and cannot be adjusted by the user.

[0131] By providing a user with a control unit comprising a means to control liquid flow rate in this way, the user is able to select a particular setting or condition of the generator which results in the generation of a liquid product having tailored properties.

[0132] Any suitable means known to the skilled person for controlling flow rates may be used. For example, the control unit may comprise an input means which allows the user to increase or decrease the flow rate. The input means may comprise, for example, a control knob or button. The control unit may comprise a display comprising an indication of the current flow settings. The control unit may comprise indicia, for example information on a digital display, which provides information on the nature of the product generated by the generator at the currently selected flow rates.

[0133] In some embodiments, the generator does not comprise any pumps at a location upstream of the cell. In some embodiments, the generator does not comprise any pumps at a location upstream of the first and second liquid outlets. In other words, the flow of liquid through the cell is entirely achieved by “pulling” liquid through the cell and out of the liquid outlets using the one or more downstream pumps, without any pump being present which “pushes” liquid through the cell from upstream. It has been found that this arrangement provides more precise control over the flow of liquid through the cell and facilitates the achievement of a flow imbalance between the first and second liquid outlets. By separately “pulling” liquid from each outlet without any “pushing”, the flow rate at each outlet can be fine-tuned, thereby permitting fine-tuning of the composition of the electrolysis products leaving the generator.

[0134] The third aspect of the invention is a method of producing hypochlorous acid using a cell according to the first aspect, the method comprising: [0135] passing brine through the brine inlet and into the electrolysis chamber; [0136] applying a voltage across the electrical connections of the anode and cathode to electrolyse the brine; [0137] drawing a first product stream from the first liquid outlet at a first flow rate, the first product stream comprising hypochlorous acid; and [0138] drawing a second product stream from the second liquid outlet at a second flow rate, the second product stream comprising sodium hydroxide; [0139] wherein the concentration of hypochlorous acid in the first product stream is higher than the concentration of hypochlorous acid in the second product stream.

[0140] In some embodiments, the second flow rate is higher than the first flow rate. In some embodiments, the second flow rate is at least 110% of the first flow rate, for example at least 115%, at least 120%, at least 125%, at least 130%, at least 135%, at least 140%, at least 145% or at least 150% of the first flow rate. In some embodiments, the second flow rate is from 110-250% of the first flow rate, for example from 110-220%, from 110-200%, from 120-200%, from 120-180%, from 120-150%, from 120-140% or from 130-140%.

[0141] In some embodiments, the second flow rate is higher than the first flow rate by at least 2 mL/min, for example at least 5 mL/min or at least 10 mL/min. In some embodiments, the second flow rate is higher than the first flow rate by 11 mL/min.

[0142] In some embodiments, the first flow rate is from 2 to 30 mL/min, for example from 3 to 25 mL/min, from 4 to 24 mL/min or from 5.5 to 22 mL/min. In some embodiments, the first flow rate is about 5.5 mL/min.

[0143] In some embodiments, the second flow rate is from 5 to 60 mL/min, for example from 6 to 55 mL/min, for example from 7 to 50 mL/min. In some embodiments, the first flow rate is about 7.7 mL/min.

[0144] In some embodiments, the method comprises varying one or more of the first flow rate and the second flow rate in order to adjust the composition or properties of the products produced at one or more of the first liquid outlet and the second liquid outlet.

[0145] In some embodiments, the method comprises fixing the first flow rate and adjusting the second flow rate in order to vary the composition of the first product stream drawn from the first liquid outlet. It has been found that adjusting the second flow rate is a convenient way to alter the pH within the cell, which affects the nature of the product dispensed from the first liquid outlet, as explained in detail above.

[0146] In some embodiments, the method comprises fixing the first flow rate at a value $F_{sub.1}$ and adjusting the second flow rate $F_{sub.2}$ in order to vary the composition of the first product stream drawn from the first liquid outlet, wherein $F_{sub.2} > F_{sub.1}$ both before and after the adjustment.

[0147] Ensuring that $F_{sub.2}$ remains higher than $F_{sub.1}$ throughout the method provides the benefit of efficient extraction of hydrogen gas bubbles from the second liquid outlet of the cell, which are formed at the cathode.

[0148] In some embodiments, before the adjustment $F_{sub.2}/F_{sub.1}$ takes a value within the range 1.1 to 2.0, and after the adjustment $F_{sub.2}/F_{sub.1}$ takes a value within the range 1.1 to 2.0, wherein the values of $F_{sub.2}/F_{sub.1}$ before and after the adjustment are different from one another. In this way, the relative flow rate is different before and after the adjustment, altering the composition of the product drawn from the first liquid outlet. In some embodiments the value of $F_{sub.1}$ is the same before and after the adjustment (i.e. the first flow rate $F_{sub.1}$ is not changed).

[0149] In some embodiments, the method comprises fixing the first flow rate $F_{sub.1}$ at a value of from 2 to 30 mL/min, and adjusting the second flow rate $F_{sub.2}$ from a first value in the range 5 to 60 mL/min to a second value in the range 5 to 60 mL/min different from the first value, wherein both the first and second value of the second flow rate $F_{sub.2}$ are greater than $F_{sub.1}$ by a factor of between 1.1 and 2.0, for example from 1.1 to 1.5, or from 1.2 to 1.4.

[0150] In some embodiments, the method comprises drawing the first product stream from the first liquid outlet at the first flow rate using a first peristaltic pump and drawing the second product stream from the second liquid outlet at the second flow rate using a second peristaltic pump. The first and second peristaltic pumps may be located downstream of the respective first and second liquid outlets, thereby arranged to “pull” liquid from each outlet.

[0151] In some embodiments, the method does not include any step of operating a pump upstream of the cell. For the reasons set out above in respect of the first aspect, there are advantages associated with “pulling” liquid through the cell rather than “pushing”.

[0152] In some embodiments, before passing through the liquid inlet the brine has an NaCl concentration of from 0.5 to 20 g/L, for example from 0.5 to 15 g/L, from 0.5 to 10 g/L or from 1 to 10 g/L.

[0153] In some embodiments, the method further comprises reversing the polarity of the anode and the cathode after a predetermined electrolysis time period. Reversal of the polarity in this way may be particularly useful when hard water is used as the basis for making the brine solution. The use of hard water will cause limescale deposits onto the electrodes during electrolysis which will reduce the electrolysis efficiency as the deposits build up. Reversing the polarity will help to remove these deposits for continued cell functioning.

[0154] In some embodiments the method comprises performing the reversal at a time interval of up to 10 mins.

[0155] In some embodiments, the ratio of operation time in the normal (product generation) mode to the operation time in the polarity switched (cleaning) mode is from 5:1 to 20:1, for example from 5:1 to 15:1, for example from 10:1 to 15:1, for example about 12:1.

[0156] In some embodiments, for each 60-minute period in the normal (product generation) mode the method comprises running the cell in the polarity switched (cleaning) mode for between 2 and 10 minutes, for example between 3 and 8 minutes, for example about 5 minutes. In some

embodiments, the cell is operated under galvanostatic conditions. In other words, the cell is operated by supplying a fixed average current to the cell. This ensures consistent electrolysis and consistent product formation. Furthermore, variations in voltage during galvanostatic operation may then be detected and can provide information on the current condition of the cell, for example whether limescale has built up causing reduced cell performance and the need for a reversal of polarity to remove limescale from the electrodes and restore cell performance.

[0157] In some embodiments, the flowrate of the brine into the liquid inlet is from 2 to 50 mL/min, for example from 5 to 50 mL/min, from 2 to 22 mL/min or from 5 to 22 mL/min.

[0158] In some embodiments, the first product stream and the second product stream are combined together in predetermined proportions to provide a third product stream. For example, the sodium hydroxide solution and hypochlorous acid solution may be combined in a predetermined proportion to provide a third product stream of predetermined pH or chemical composition thereby providing a product which may be tailored for a specific end use.

[0159] In some embodiments, the method further comprises diluting one or more of the first product stream and the second product stream with water to provide a diluted product stream, wherein the diluted product stream has an NaCl concentration of less than 1 g/L.

[0160] In some embodiments, the cell is at least partially submerged in a brine reservoir which contains the brine, and the method comprises drawing brine into the cell through the brine inlet from the brine reservoir.

[0161] In other embodiments, brine is fed to the brine inlet of the cell through a brine feed conduit. The brine feed conduit may be connected with the brine inlet of the cell and adapted to feed brine from a brine reservoir to the cell. The brine feed conduit may comprise a tube, for example a flexible tube. The brine feed conduit may comprise a flexible plastic tube.

[0162] In some embodiments, the cell is operated at a current within the range 0.4 to 1.5 A, for example from 0.5 to 1.2 A, from 0.6 to 1.0 A, from 0.7 to 0.9 A, or at about 0.8 A.

[0163] In some embodiments, the cell is operated at a voltage within the range 5 to 20 V, for example from 6 to 18 V, from 6 to 15 V, from 6 to 12 V or from 7 to 9 V.

[0164] In some embodiments, the cell is operated at a current within the range 0.4 to 1.5 A and at a voltage within the range 5 to 20 V.

[0165] In some embodiments the cell is operated at a current of about 0.8 A and a voltage of about 7-9V.

[0166] In some embodiments, the voltage is delivered from a battery, for example a rechargeable battery such as a lithium-ion battery, via a pulse width modulation (PWM) module.

[0167] In some embodiments, the pH within the cell during electrolysis is maintained within the range 4.5 to 9.5.

[0168] In some embodiments, the method comprises adjusting the pH within the cell by adjusting one or more of the first and second flow rates.

[0169] The fourth aspect of the invention is a household cleaning product dispenser comprising the electrolytic cell according to the first aspect. The dispenser comprises suitable dispensing means. Any suitable dispensing means known to the skilled person may be used, for example a spray nozzle, such as a pump action spray nozzle comprising a mechanical actuator. The mechanical actuator may be a trigger for actuation by the user to dispense the cleaning product.

[0170] In some embodiments, the household cleaning product dispenser comprises control means to adjust the composition of the product to be dispensed. In some embodiments, the control means may comprise an adjustable input adapted to vary one or more of the first flow rate and second flow rate, thereby adjusting the properties of the product, for example adjusting the product pH. In this way, the user of the household cleaning product dispenser is provided with control over the properties of the product dispensed and is able to tailor the properties of the product to an intended use.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0171] FIG. 1 is a perspective cutaway view of an electrolytic cell according to a first embodiment of the invention.

[0172] FIG. 2 is (a) a top view of an electrolytic cell according to the first embodiment of the invention, and (b) a bottom view of an electrolytic cell according to the first embodiment of the invention.

[0173] FIG. 3 is a side view of an electrolytic cell according to the first embodiment of the invention.

[0174] FIG. 4 is a schematic diagram showing the unit operations of a hypochlorous acid generator according to an embodiment of the invention.

[0175] FIG. 5 is a plot of the pH and HOCl output measured over a range of outlet flow biases.

DETAILED DESCRIPTION

[0176] An exemplary embodiment of the electrolytic cell 1 of the invention is shown in FIG. 1, which shows a perspective view of the cell in part-cutaway form.

[0177] The cell includes an anode 101 and a cathode 102 contained within an outer housing or casing 103 which defines an electrolysis chamber 104. The electrolysis chamber contains the anode 101 and cathode 102 with a diaphragm 105 disposed between the anode and cathode.

[0178] The housing 103 is made from acrylic and the wall of the housing has a thickness of 3-12 mm. The width of the housing is 58 mm and the total height of the housing is 80 mm.

[0179] Each of the anode 101 and cathode 102 is made from titanium carrying an IrO.sub.2/RuO.sub.2 mixed metal oxide catalyst coating. Each of the anode 102 and cathode 102 is a mesh including a plurality of apertures 106, 107.

[0180] The diaphragm 105 is made from lint free cloth and has a thickness of 0.3 mm.

[0181] The separation between the anode 101 and the cathode 102 is 3 mm. The anode and cathode lie parallel to one another within the cell.

[0182] The presence of the anode 101 and cathode 102 within the cell divides the interior cell volume to define an inlet chamber 108 between the anode 101 and cathode 102, and containing the diaphragm 105, and two collection chambers: an anode collection chamber 109 between the anode 101 and the internal wall of the housing 103 and a cathode collection chamber 110 between the cathode 102 and the internal wall of the housing 103.

[0183] A first liquid outlet 111 is provided in the wall of the housing 103 proximal the anode 101.

[0184] The first liquid outlet 111 consists of a short conduit 112 having an opening 113 within the wall of the housing 103. The conduit leads from the opening out of the housing for the passage of liquid out of the cell. The conduit extends as a short stem which terminates at an open end 114 which is adapted for connection to a pipe, for example flexible tubing, to feed the liquid to a storage tank or dispenser (not shown). The stem of the conduit 112 has a bevelled annulus at the end around the opening 114 to facilitate connection with such a pipe.

[0185] A second liquid outlet 115 is provided in the wall of the housing 103 proximal the cathode 102. The second liquid outlet 115, like the first liquid outlet, consists of a short conduit having an opening within the wall of the housing 103. The conduit leads from the opening out of the housing for the passage of liquid out of the cell. The conduit extends as a short stem which terminates at an open end which is adapted for connection to a pipe, for example flexible tubing, to feed the liquid to a storage tank or dispenser (not shown). The stem of the conduit has a bevelled annulus at the end around the opening to facilitate connection with such a pipe (note that for clarity the features of the second liquid outlet which are analogous to the features of the first liquid outlet already discussed above are not labelled in FIG. 1).

[0186] A first peristaltic pump head (not shown) is associated with the tubing which is connected to

the short conduit **112** of the first liquid outlet **111**. The operation of the first peristaltic pump head, provided by the action of a separate motor (not shown) which is powered by a battery power source via a PWM module (not shown), pumps liquid through the tubing away from the cell.

[0187] Analogously, a second peristaltic pump head (not shown) is associated with the tubing which is connected to the second liquid outlet **115**. The operation of the second peristaltic pump head, provided by the action of a separate motor (not shown) which is powered by a battery power source via a PWM module (not shown), pumps liquid through the tubing away from the cell.

[0188] The first and second peristaltic pump heads are each powered by the same battery via the same motor and PWM module.

[0189] The cell also includes a first electrical connection **116** which extends from the housing **103** and is electrically connected to the anode **101**. Similarly, a second electrical connection **117** which extends from the housing **103** is electrically connected to the cathode **102**. The connections **116** and **117** can be connected to an external power supply (not shown) such as a battery to apply a voltage across the anode and the cathode to perform electrolysis of a solution within the electrolysis chamber **104**.

[0190] The same battery which is used to power the first and second peristaltic pump heads via the motor is also used to power the electrolysis by delivering a voltage across the cell.

[0191] It will be understood by the skilled person that although the above explanation identifies a first of the electrodes **101** as the anode and a second of the electrodes **102** as the cathode, the identities of the electrodes as anode or cathode will in practice depend on the polarity of the voltage applied across the electrical connections **115** and **116**. The voltage applied across the connections could of course be reversed, in which case electrode **102** would become the anode and electrode **101** would become the cathode. Such a reversal may be useful for example to place the cell in a cleaning condition, since reversal of the polarity will help to remove unwanted deposits from the electrodes which may have built up during electrolysis, especially if hard water is used as the source of water fed into the cell.

[0192] It is evident from FIG. **1** that a portion **118** of the wall of the housing **103** which is close to the first liquid outlet **111** is angled towards the first liquid outlet **111**. This provides a funnel-like structure of the internal wall of the housing **103** which funnels or narrows towards the opening **113** in the first liquid outlet **111**. This facilitates the removal of any gas bubbles from the cell via the first liquid outlet.

[0193] Similarly, a portion **119** of the wall of the housing **103** which is close to the second liquid outlet **115** is angled towards the second liquid outlet **115**. This provides a funnel-like structure of the internal wall of the housing **103** which funnels or narrows towards the opening in the second liquid outlet **115**. This facilitates the removal of any gas bubbles from the cell via the second liquid outlet.

[0194] The housing **103** of the cell defines an opening **120** at the base of the housing which provides fluid communication between the external surroundings of the housing **103** and a brine inlet (not shown) located within an inner wall of the housing **103** at the base of the cell. The location of the opening **120** is best seen in FIG. **2(b)**. The opening **120** therefore permits the flow of liquid (brine) into the cell when the housing **103** is placed within a liquid reservoir (for example a brine reservoir).

[0195] FIG. **2(a)** shows a top view of the cell **1**, in which first and second liquid outlets **111**, **115** and first and second electrical connections **116**, **117** can be easily seen.

[0196] FIG. **2(b)** shows a top view of the cell **1**, in which the opening **120** in the cell can be seen. FIG. **3** shows a side view of the cell **1**.

[0197] The cell may be used for the electrolysis of a brine solution (a sodium chloride solution). To use the cell in this way, a brine reservoir (not shown) is prepared by dissolving sodium chloride in water. This may be done by the consumer, on demand, for example at home or in the workplace. The brine reservoir may be an open-top tank or vessel which is at least partially filled with brine

prior to use of the electrolytic cell and which is dimensioned to facilitate the insertion and holding in an upright position of the housing **103** of the cell **1**.

[0198] When the housing **103** is placed in such a reservoir and submerged in such a way that brine within the reservoir enters and covers the opening **120** at the base of the housing **103**, the drawing of liquid from the first and/or second liquid outlets **111**, **115** will initiate the flow of liquid through the cell, drawing liquid into the cell through the opening **120** in the base of the housing **103**.

[0199] As brine enters the opening **120** it will pass through the brine inlet (not shown) into the central inlet chamber **108** portion of the electrolysis chamber **104**. A first portion of the brine will pass into a first half of the inlet chamber **108** on the side of the diaphragm **105** which is proximal the anode **101** and a second portion of the brine will pass into a second half of the inlet chamber **108** on the side of the diaphragm **105** which is proximal the cathode **102**. The flow conditions established by the operation of the peristaltic pumps (not shown) cause the first portion of brine to continue to flow towards and ultimately through the mesh structure of the anode **101** and into the anode collection chamber **109**. Similarly, the second portion of brine will continue to flow towards and ultimately through the mesh structure of the cathode **102** and into the cathode collection chamber **110**. At a suitable time following the initiation of the peristaltic pumps and the start of liquid flow through the cell, a suitable voltage will be applied across the electrical connections **116**, **117** from a battery power source (not shown) via a PWM module (not shown). As the brine comes into contact with the anode **101**, solvated chloride ions within the solution will be oxidised to form chlorine which in turn will react with water to form hypochlorous acid. As the brine comes into contact with the cathode **102**, water molecules within the solution will be reduced to form solvated hydroxide ions.

[0200] In this way the anode collection chamber **109** will contain a solution with a relatively high concentration of hypochlorous acid and the cathode collection chamber **110** will contain a solution with a relatively high concentration of sodium hydroxide.

[0201] The continued operation of the peristaltic pumps (not shown) will draw the hypochlorous acid solution out of the cell through the first liquid outlet **111** and into a suitable conduit (not shown) which conveys the hypochlorous acid solution to a suitable container or dispenser (not shown). The sodium hydroxide solution will be drawn out of the cell through the second liquid outlet **115** and into a suitable conduit (not shown) which conveys the sodium hydroxide solution to a suitable container or dispenser (not shown).

[0202] During operation, the liquid flow rate provided by the second peristaltic pump head is generally larger than the liquid flow rate provided by the first peristaltic pump head, such that liquid flows more quickly out of the second liquid outlet **115** than out of the first liquid outlet **111**. This facilitates the extraction of bubbles of hydrogen gas from the vicinity of the cathode and also establishes a pH within the cell which favours the production of hypochlorous acid at the anode.

[0203] After the polarity reversal discussed above, performed for example to clean the electrodes, it will be understood that the identity of the electrodes will be reversed and similarly, the identities of the first and second liquid outlets will also be reversed. So, after the polarity reversal the flow rates will be adjusted to ensure that liquid still flows more quickly out of the second liquid outlet (which was previously the first liquid outlet before reversal).

[0204] The user of the cell is able to adjust the flow rates out of the first and second liquid outlets **111**, **115**. By providing the user with the ability to adjust the flow rates, it is possible for the user to tailor the properties of the products which are generated by the cell.

[0205] The cell **1** may form part of a hypochlorous acid generator. A schematic diagram of an embodiment of a hypochlorous acid generator which includes the cell **1** is depicted in FIG. **4**.

[0206] Brine is stored in a brine reservoir **2** which may be a simple vessel such as a beaker into which brine is fed or poured. A user, for example a consumer using the generator at home, may prepare a brine solution by adding table salt to tap water and mixing to dissolve and form a homogenous solution. The brine is then fed into the cell **1**. The brine may be fed through a tube, or

alternatively the cell may simply sit in the brine reservoir allowing the brine to enter an inlet which feeds the cell. A first peristaltic pump head **3** pumps a product solution which is rich in hypochlorous acid out of the cell. A second peristaltic pump head **4** pumps a product solution which is rich in sodium hydroxide out of the cell. Both peristaltic pump heads are driven by a motor **5** which is powered by a battery pack **8**, for example a rechargeable lithium-ion battery pack, via a pulse width modulation (PWM) module **11**. The batter pack **8** also supplies a voltage to the cell **1** via a second PWM module **9**, to establish a voltage across the electrodes of the cell and drive electrolysis.

[0207] Hypochlorous acid solution from the cell is pumped by peristaltic pump head **3** into a storage container **6**. A third peristaltic pump head **10** which is also driven by the motor **5** can be used to pump brine from the brine reservoir **2** directly into the storage container **6** in order to dilute the stored hypochlorous acid solution as needed depending on the application.

[0208] Sodium hydroxide solution from the cell is pumped by peristaltic pump head **4** into a storage container **7**.

[0209] Hypochlorous acid solution in the storage container **6** and sodium hydroxide solution in the storage container **7** may be dispensed, for example using a suitable dispenser such as a pump action spray and used as needed for sanitising or cleaning purposes.

EXAMPLES

[0210] The electrolytic cell shown in FIGS. **1-3** was fed with brine solution and electrolysis was performed under the process conditions indicated in Table 1.

[0211] The anodic flow rate was fixed as indicated in Table 1, and the cathodic flow rate was varied.

[0212] When the cell was tested using brine solutions made with very hard water (a hardness of around 300 ppm calcium), the polarity of the cell was switched at least every 10 minutes during the electrolysis operation in order to remove limescale debris from the electrodes.

[0213] In each test run, electrolysis was performed for >200 hours.

TABLE-US-00001

TABLE 1	Example 1	Example 2	Example 3	Example 4	Brine NaCl	4	4	4	4
concentration/g/L	5.5	5.5	5.5	5.5	5.5	6.5	7.5	8.5	
Anodic flow rate/mL/min	5.5	5.5	5.5	5.5	5.5	6.5	7.5	8.5	
Cathodic flow rate/mL/min	5.5	6.5	7.5	8.5					
Current/A	0.8	0.8	0.8	0.8					

[0214] FIG. **5** shows the results of the electrolysis runs of Examples 1-4. The variation in HOCl output and pH is indicated for the three different cathodic flow rates at fixed anodic flow rate of 5.5 mL/min.

Example 1

[0215] At the balanced flow rates of Example 1 (5.5 mL/min at the cathode and 5.5 mL/min at the anode), the HOCl output was relatively low. Without wishing to be bound by theory, it is believed that the lower cathode flow means that the vicinity of the cathode is not being fully liberated of bubbles of H₂ which are formed there. This causes more NaOH to bleed across the cell, increasing the pH slightly thereby increasing the amount of active chlorine at the anode which is in the form of sodium hypochlorite rather than hypochlorous acid. The close-to-neutral pH (pH 7.4) of this solution nevertheless provides a benefit of being more suitable on delicate surfaces which may be damaged by lower pH, and would still have disinfectant properties, but the disinfecting power of the solution will be lower because of the relatively low level of active chlorine in HOCl form.

Example 2

[0216] When the cathode flow is increased to 6.5 mL/min the amount of HOCl generated at the cathode increases. The pH is similar to Example 1. The slightly higher cathode flow removes more H₂ gas from the cell, increasing the amount of HOCl generated at the anode relative to Example 1.

Example 3

[0217] At a moderate flow imbalance in favour of the cathode (7.5 mL/min at the cathode and 5.5 mL/min at the anode), the amount of HOCl generated at the cathode is the highest observed, at

around 9300 ppm.Math.mL/min. The pH is slightly lower than observed in Example 1 (around pH 6), further favouring the formation of HOCl at the anode. Without wishing to be bound by theory, it is believed that this flow imbalance is sufficient to prevent a build-up of H.sub.2 gas bubbles at the cathode, such that pH at the anode is low enough to favour HOCl formation. However the flow imbalance is not great enough to cause any significant bleed of HOCl across to the cathode, such that the HOCl output at the anode remains high.

Example 4

[0218] At very high flow imbalance in favour of the cathode (8.5 mL/min at the cathode and 5.5 mL/min at the anode), the HOCl generated begins to bleed over into the cathode flow, reducing the amount of HOCl generated. The high cathode flow rate means that more H.sub.2 gas is removed with the cathode flow, reducing the pH of the anode solution to around pH 1. Such a solution may find use in applications where a more acidic hypochlorous acid solution is required, and would still have disinfectant properties, but the disinfecting power of the solution will be lower because of the relatively low level of active chlorine in HOCl form. The product may be useful in, for example, the sterilising of robust surfaces, due to the high concentration of dissolved chlorine which would be present as such low pH. The product may also be used after downstream dilution, again due to the high chlorine concentration.

[0219] Table 2 sets out typical operating conditions for electrolytic cells according to the invention.

TABLE-US-00002

TABLE 2	Operating Conditions
Flowrate	5.5-22 mL/min through each electrode
Current	0.4-0.8 A
Voltage	5 to 20 V
Power supply	Galvanostatic; PWM used with a battery pack power source
Water parameters	Sodium Chloride 1-6 g/L concentration
Max Temperature	40° C.
Water hardness	300 mg/L.sub.CaCO3
Inlet pH value	5-8
Electrode Stack	Electrodes Ruthenium/Iridium Mixed Metal Oxide coated titanium
Diaphragm	Porous material (e.g. lint-free cloth)
Electrode	Approx. 45 × 17 mm dimension
Connections and Miscellaneous	Electrical Stranded wire or banana plug or spring connector connections
Water Tube connection	connections
Cell dimensions	60 × 82 × 27 mm

Claims

1. An electrolytic cell for the production of hypochlorous acid and sodium hydroxide from brine as separate product streams, the electrolytic cell comprising: an electrolysis chamber comprising an anode and a cathode; electrical connections to each of the anode and cathode; a brine inlet into the electrolysis chamber for supplying a brine solution into the electrolysis chamber; a first liquid outlet from the electrolysis chamber proximal the anode; a second liquid outlet from the electrolysis chamber proximal the cathode; and at least one pump arranged to draw liquid from the electrolysis chamber through the first liquid outlet at a first flow rate and through the second liquid outlet at a second flow rate; wherein either: (a) the first flow rate and second flow rate are variable relative to one another; or (b) the first flow rate and second flow rate are fixed relative to one another at a predetermined difference in flow rate.
2. An electrolytic cell according to claim 1 wherein the at least one pump comprises: a first pump arranged to draw liquid from the electrolysis chamber through the first liquid outlet at the first flow rate; and a second pump arranged to draw liquid from the electrolysis chamber through the second liquid outlet at the second flow rate.
3. (canceled)
4. (canceled)
5. An electrolytic cell according to claim 1, wherein; (i) the anode and the first liquid outlet are arranged such that at least a portion of the liquid passing from the brine inlet to the first liquid outlet must pass through the anode; and/or the cathode and the second liquid outlet are arranged such that at least a portion of the liquid passing from the brine inlet to the second liquid outlet must pass through the cathode; or (ii) the anode divides the electrolysis chamber into an inlet chamber on

a first side of the anode, adjacent the cathode, and an anode collection chamber on a second side of the anode, wherein the first liquid outlet is located in the anode collection chamber; such that, in use, liquid passes from the brine inlet into the inlet chamber, then through the anode into the anode collection chamber before passing out of the first liquid outlet; or (iii) the cathode divides the electrolysis chamber into an inlet chamber on a first side of the cathode, adjacent the anode, and a cathode collection chamber on a second side of the cathode, wherein the second liquid outlet is located in the cathode collection chamber; such that, in use, liquid passes from the brine inlet into the inlet chamber, then through the cathode into the cathode collection chamber before passing out of the second liquid outlet; or (iv) each of the anode and cathode are liquid-impermeable and the electrolytic cell comprises means for the passage of liquid past or around the electrode; or (v) one or both of the anode and cathode comprise a metal substrate and a noble metal oxide catalyst supported on the substrate; or (vi) each of the anode and cathode are liquid-permeable.

6. An electrolytic cell according to claim 1, further comprising a housing defining the electrolysis chamber and containing the anode and cathode, the housing comprising an opening in fluid communication with the brine inlet and the electrolysis chamber, such that when the housing is at least partially submerged in a brine reservoir, brine from the reservoir enters the housing through the opening and is drawn through the electrolytic cell by operation of the at least one pump to draw liquid from the electrolysis chamber.

7. An electrolytic cell according to claim 1, comprising a porous diaphragm arranged between the anode and the cathode.

8. An electrolytic cell according to claim 7, wherein the porous diaphragm is a lint free cloth.

9. An electrolytic cell according to claim 1, wherein the electrolytic cell does not contain any diaphragm structure between the anode and the cathode.

10. An electrolytic cell according to claim 1, wherein the electrolytic cell is configured to provide a higher liquid flow rate from the second liquid outlet than the first liquid outlet.

11. (canceled)

12. (canceled)

13. (canceled)

14. A hypochlorous acid generator comprising an electrolytic cell according to claim 1, further comprising: one or more of a brine reservoir and a brine feed conduit; a hypochlorous acid storage unit; a sodium hydroxide storage unit; a power source; and a product dispensing device.

15. A method of producing hypochlorous acid using an electrolytic cell according to claim 1, the method comprising: passing brine through the brine inlet and into the electrolysis chamber; applying a voltage across the electrical connections of the anode and cathode to electrolyse the brine; drawing a first product stream from the first liquid outlet at a first flow rate, the first product stream comprising hypochlorous acid; and drawing a second product stream from the second liquid outlet at a second flow rate, the second product stream comprising sodium hydroxide; wherein the concentration of hypochlorous acid in the first product stream is higher than the concentration of hypochlorous acid in the second product stream.

16. A method according to claim 15, wherein the second flow rate is higher than the first flow rate.

17. A method according to claim 15, wherein before passing through the liquid inlet the brine has an NaCl concentration of from 0.5 to 20 g/L.

18. A method according to claim 15, further comprising reversing the polarity of the anode and the cathode after a predetermined electrolysis time period.

19. A method according to claim 15, wherein the electrolytic cell is operated under galvanostatic conditions.

20. A method according to claim 15, wherein the flowrate of the brine into the liquid inlet is from 2 to 50 mL/min.

21. A method according to claim 15, wherein the first product stream and the second product stream are combined together in predetermined proportions to provide a third product stream.

22. A method according to claim 15, further comprising diluting one or more of the first product stream and the second product stream with water to provide a diluted product stream, wherein the diluted product stream has an NaCl concentration of less than 1 g/L.
23. A method according to claim 15, wherein: (i) the electrolytic cell is at least partially submerged in a brine reservoir which contains the brine, and the method comprises drawing brine into the electrolytic cell through the brine inlet from the brine reservoir; or (ii) the electrolytic cell is operated at a current within the range 0.4 to 0.8 A and at a voltage within the range 5 to 20 V.
24. (canceled)
25. A method according to claim 15, wherein the pH within the electrolytic cell during electrolysis is maintained within the range 4.5 to 9.5.
26. A household cleaning product dispenser comprising the electrolytic cell according to claim 1.
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